

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 02

Miscellaneous Topics



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Recap of Previous Lecture



Number System



Topics to be Covered

Number System Cont.



#Q. A linear equation in x -base system is given as:

$$\begin{cases} 2x + 3y = 15 \\ 3x + 2y = 14 \end{cases} \quad x \geq 6$$

If one of the solution of this equation is $x = (2)_x$
 $= (2)_{10}$
then value of x 8.

$$(2)_x = (2)_{10} \quad (15)_x = (x+5)_{10}$$

$$(3)_x = (3)_{10} \quad (14)_x = (x+4)_{10}$$

$$(13)_{10} = (15)_8$$

$$[2x + 3y = x+5]_{10}$$

$$[3x + 2y = x+4]_{10}$$

$$4 + 3y = x+5 \quad \times 2$$

$$6 + 2y = x+4 \quad \times 3$$

$$10 = 3(x+4) - 2(x+5) = x+2$$

$$\boxed{x=8}$$

$$\rightarrow 4 + 3y = 13$$

$$3y = 9$$

$$y = (3)_{10}$$

$$= (3)_8$$

Q. A quadratic equation mentioned in two different no system is given below:

a. $(x^2 - \underline{14}x + 43)_{n_1} = (x^2 - 11x + 32)_{n_2}$

Then value of $(n_1 + n_2)$ 19.

b. $(x^2 + 5x - 124)_{n_1} = (x^2 + 11x - 1110)_{n_2}$

then $(n_1 + n_2)$ 12.

$$(5)_{n_1} = (11)_{n_2} \rightarrow (5)_{10} = (n_2 + 1)_{10} \rightarrow n_2 = (4)_{10}$$

$$(124)_{n_1} = (1110)_{n_2}$$

$$(n_1^2 + 2n_1 + 4)_{10} = n_2^3 + n_2^2 + n_2 + 0$$

$$n_1^2 + 2n_1 + 4 = 84$$

$$n_1^2 + 2n_1 - 80 = 0$$

$$\begin{aligned} (14)_{n_1} &= (11)_{n_2} \\ (43)_{n_1} &= (32)_{n_2} \end{aligned}$$

$$(n_1 + 4)_{10} = (n_2 + 1)_{10}$$

$$(4n_1 + 3)_{10} = (3n_2 + 2)_{10}$$

$$n_2 - n_1 = 3$$

$$3n_2 - 4n_1 = 1$$

$$\boxed{n_2 = 11} \quad \boxed{n_1 = 8}$$

$$x_1^2 + 2x_1 - 80 = 0$$

$$(x_1 + 10)(x_1 - 8) = 0$$

$$x_1 = -10 \times, \quad 8 \checkmark \checkmark$$

$$x_2 = 4$$



Q. A relationship b/w two no.s is as given below:

$$(15)_n = (13)_{n_1}$$

$$n_1 > n$$

then minimum value of $(n+n_1)$ 14.

$$\begin{aligned} n &\geq 6 \\ &\& \\ n_1 &\geq 4 \end{aligned}$$

$$\begin{aligned} (15)_n &= (n+5)_{10} \\ (13)_{n_1} &= (n_1+3)_{10} \end{aligned}$$

$$(n+5)^2 = (n_1+3)^2$$

$$\begin{aligned} n+5 &= \pm \sqrt{(n_1+3)^2} = \pm |n_1+3| \\ &= \pm (n_1+3) \end{aligned}$$

$$n+5 = (n_1+3)$$

$$n = (n_1 - 2) \rightarrow n_1 = (n + 2)$$

$$\begin{aligned} (8, 6) & (9, 7) \dots (10, 8) \dots \\ \downarrow \quad \downarrow \\ n_1 \quad n \end{aligned}$$

$$\sqrt{a^2} = |a|$$

a - real number

$$x^2 = 4$$

$$x = \pm \sqrt{4}$$

$$|2| = 2$$

$$|-2| = -(-2)$$

$$|b| = b \quad \text{if } b = +ve$$

$$|b| = -b \quad \text{if } b = -ve$$

#Q. A relationship b/w two no. is as given below:

$$(62)_x = (y2)_9$$

Where x and y are +ve integers.

Then minimum value of $(x+y)$ 15.

Maximum value of $(x+y)$ 20.

No. of possible solutions of this equation are 2.

$$\rightarrow x \geq 7$$

&

$$0 \leq y \leq 8$$

$$(6x+2)_{10} = (9y+2)$$

$$6x = 9y$$

$$x = \frac{3}{2}y$$

$$y=6 \rightarrow x=9$$

$$y=8 \rightarrow x=12$$

• b. $(63)_x = (y9)_{12}$

$$(x+y)_{\min} \underline{10}$$

$$(x+y)_{\max} \underline{34}$$

No. of solutions of equation 9.

$$x \geq 7$$

&

$$0 \leq y \leq 11$$

$$(6x+3)_{10} = 12y+9$$

$$x = 2y+1$$

$$y=3, x=7$$

$$3 \leq y \leq 11 \rightarrow x=7, 9, 11, \dots, 23$$



#Q. $(xzy)_4 = (52)_3$

$$0 \leq x \leq 3, \text{ \& } 0 \leq y \leq 3$$

$$\text{ \& } z \geq 6$$

Minimum value of $(x+y+z)$ $\frac{12}{15}$

Maximum " " " " $\frac{15}{15}$

No. of solutions of given equation 2

$$(16x + 8 + y)_{10} = (53 + 2)_{10}$$

$$53 = 16x + y + 6$$

$$3 = \frac{16x + y + 1}{5} + 1$$

$$x=0 \rightarrow \text{No solution.}$$

$$x=1 \rightarrow \text{No solution}$$

$$x=2, y=2 \rightarrow z=8 - \text{Valid solution}$$

$$x=3, y=1 \rightarrow z=11 \rightarrow \text{Valid solution.}$$

#Q. A relationship b/w two nos is as given below:

$$\sqrt{(169)_x} = (13)_x \quad x \geq 10$$

then which of the following option is correct

a. $x = 10$

b. $x \geq 10$

c. $x \leq 9$

d. None of them

$$\sqrt{(x^2 + 6x + 9)_{10}} = (x+3)_{10}$$

$$x^2 + 6x + 9 = (x+3)^2$$

$$x^2 + 6x + 9 = x^2 + 6x + 9$$

↓ satisfying for any value of x

overall $\rightarrow (x \geq 10)$

#Q. A relationship b/w two nos is as given below:

$$\sqrt{(225)x} = (15)x \quad \checkmark \quad x \geq 6$$

then which of the following option is correct

☒ a. $x = 10$

$$\sqrt{(2x^2 + 2x + 5)}_{10} = (x+5)_{10}$$

b. $x \geq 10$

c. $x \leq 9$

d. $x \geq 6$

$$2x^2 + 2x + 5 = (x+5)^2$$

$$2x^2 + 2x + 5 = x^2 + 10x + 25$$

$$x^2 - 8x - 20 = 0$$

$$(x-10)(x+2) = 0$$

$$x = 10, -2$$

☒ $x = 10$ X

A quadratic equation in base n -system is as given below:

$$(x^2 - 16x + 51)_n = 0 \quad n > 7$$

If one of the root of this equation is $x = (4)_n = (4)_{10}$

then value of n 7.

Value of the other root $(12)_n$, $(9)_{10}$

$$(16)_n = (n+6)_{10}$$

$$(51)_n = (5n+1)_{10}$$

$$[x^2 - (n+6)x + (5n+1)]_{10} = 0$$

$$x = (4)_{10} \rightarrow 16 - (n+6)4 + 5n+1 = 0$$

$$n - 7 = 0$$

$$n = 7$$

$$(x^2 - 13x + 36)_{10} = 0$$

$$(x-9)(x-4) = 0$$

$$x = (4), (9)_{10}$$

$$= (4)_7, (12)_7$$

$$(13)_{10} = (16)_7$$

$$(36)_{10} = (51)_7$$

$$\alpha + \beta = 16$$

$$(4)_n + (\beta)_n = (16)_n$$

$$(\beta)_n = (16)_n - (4)_n$$

$$(11)_n - (4)_n = (12)_n$$

$$= (n-3)_n$$

[Concept of Compliments]



$$\text{base} = r$$

→ $(r-1)$'s complement = [highest no. of length n in r base system]
— number of length n .

→ r 's complement = $(r-1)$'s complement + 1

→

$(786)_{10}$

$9's$
 $\rightarrow$

$10's$
 $\rightarrow$

$$\begin{array}{r} 999 \\ - 786 \\ \hline (213)_{10} \end{array}$$

$$(214)_{10}$$

$$\begin{aligned} (2347890)_{10} &\xrightarrow{9's} (7652109)_{10} \\ &\xrightarrow{10's} (7652110)_{10} \end{aligned}$$

• $(546013)_{10}$

├── 9's
└── 10's

$(453986)_{10}$ $(453987)_{10}$

Lets understand with examples $\rightarrow$ 1's and 2's compliments

• $(1011)_2$

$1's$	$\rightarrow$	1111
		1011
		$(0100)_2$
$2's$	$\rightarrow$	$(0101)_2$

$$(10101101)_2 \xrightarrow{2's} (01010011)_2$$

$$(10100)_2 \xrightarrow{2's} (01100)_2$$

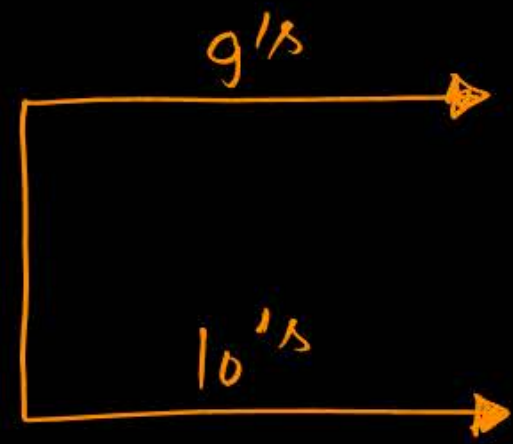
$$(110101000)_2 \xrightarrow{2's} (001011000)_2$$

• $(10101101)_2$

$1's$	$\rightarrow$	$(01010010)_2$
$2's$	$\rightarrow$	$(01010011)_2$



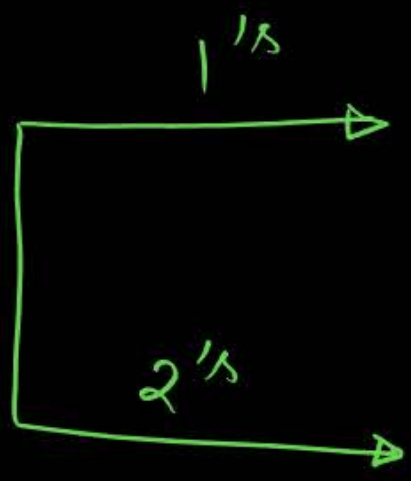
• $(97.86)_{10}$



$$\begin{array}{r} 9999 \\ 9786 \\ \hline 02.13 \\ \\ 0213 \\ 1 \\ \hline 02.14 \\ \hline \end{array}$$

02.13 ✓✓
 03.13 ✗

• $(110.101)_2$



$$(001.010)_2$$

$$010.010 \text{ ✗}$$

$$(001.011)_2 \text{ ✓✓}$$



• $(0000)_{10} \xrightarrow{9's} (9999)_{10}$
 $\xrightarrow{10's} \begin{array}{r} 9999 \\ + 1 \\ \hline \end{array}$

$$(0000)_9 \xrightarrow{9's} (0000)_9$$

• $(0000)_2 \xrightarrow{1's} (1111)_2$
 $\xrightarrow{2's} \begin{array}{r} 1111 \\ + 1 \\ \hline \end{array}$
 $\xrightarrow{\text{discard } 10000}$

5-bit numbers

$$(00000)_2 \xrightarrow{2's} (00000)_2$$
$$(00000)_2 \xrightarrow{2's} (10000)_2$$

$$(10000)_2 \xrightarrow{2's} (10000)_2$$

$$\left[\underbrace{10000 \dots 0}_{(n-1) \text{ times}} \right]_2 \xrightarrow{2's} \left[1 \underbrace{0000 \dots 0}_{(n-1) \text{ times}} \right]_2$$

$$1000 \xrightarrow{2's} (1000)_2$$
$$10000 \xrightarrow{2's} (10000)_2$$

- $$(1011)_2 \xrightarrow{1's} (0100)_2 \xrightarrow{1's} (1011)_2$$

$$\quad \quad \quad \curvearrowright \quad \quad \quad \uparrow$$

$$\quad \quad \quad 1's$$

$$(A)_2 \xrightarrow{1's} (B)_2$$

$$\quad \quad \quad \curvearrowright$$

$$\quad \quad \quad 1's$$

- $$1011 \xrightarrow{2's} (0101)_2 \xrightarrow{2's} (1011)_2$$

$$(A)_2 \xrightarrow{2's} (B)_2 \xrightarrow{2's} (A)_2$$

$$\quad \quad \quad \curvearrowright$$

$$\quad \quad \quad 2's$$



Topic : 2 Min Summary

→ Number system.



Thank you

GW
Soldiers !

